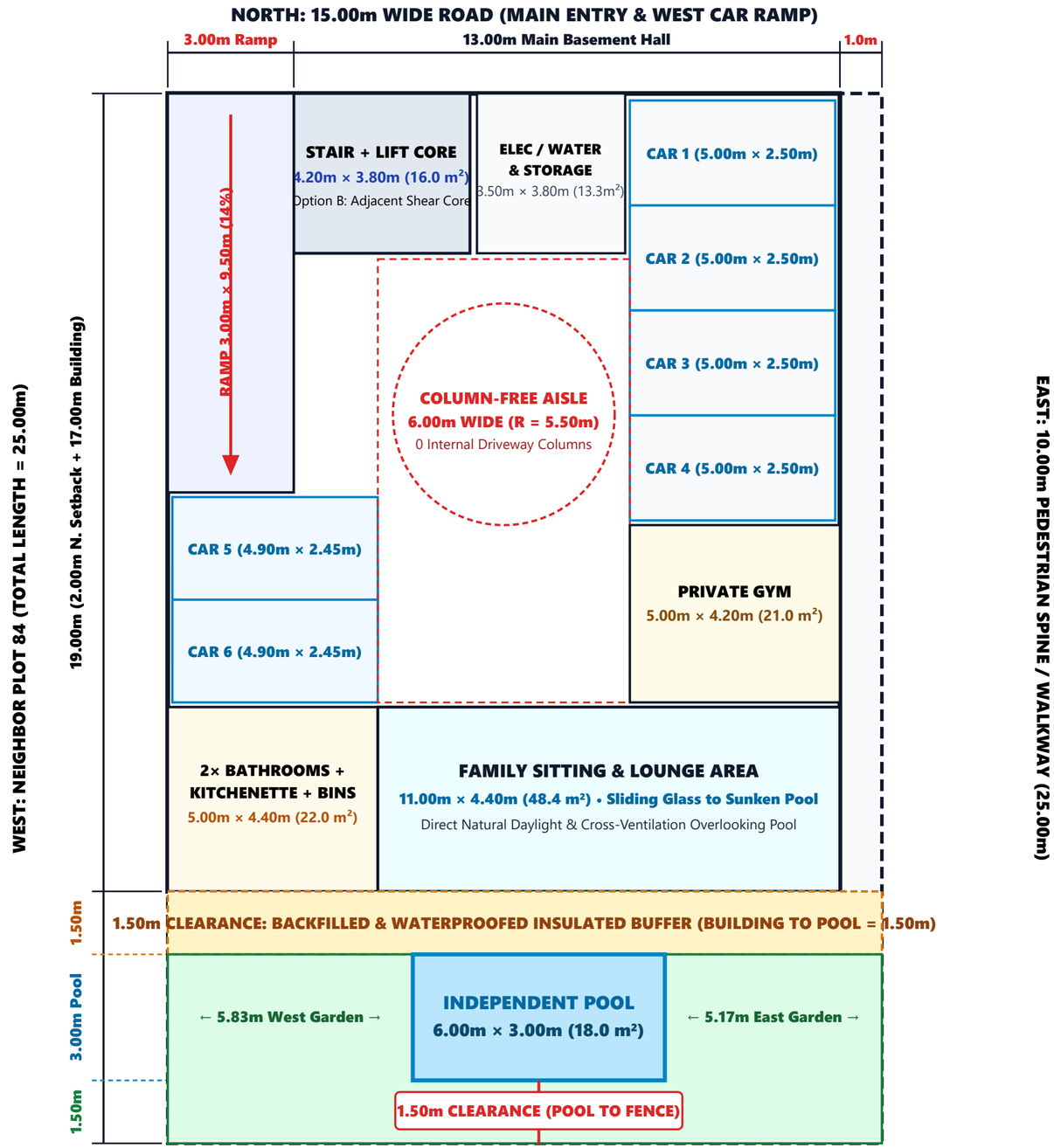
SHEET A-101: BASEMENT LEVEL PLAN (-3.20m) & REAR POOL 1.50m CLEARANCES

Plot 83 (17.00m E–W × 25.00m N–S = 425.00 m²) • 6-Car Column-Free Garage + Family Lounge + Independent 6.00m × 3.00m Pool

REV 2 — UPDATED PER CLIENT FEEDBACK



REVISION 2 UPDATES (PER CLIENT FEEDBACK)

- **Pool Clearances Updated:** Exactly **1.50m clearance** between Building South Wall and Pool (insulated & backfilled buffer) AND **1.50m clearance** between Pool and Southern Fence (total rear garden depth = `1.50m + 3.00m + 1.50m = 6.00m`).
- **Full Room Dimensions Added:** Every space, stall, ramp, aisle, and setback now displays exact `Width × Length (m)` in addition to `Area (m²)`.
- **High-Clarity Large Format:** Delivered as 4 individual large-scale sheets + multi-page vector PDF + AutoCAD `.DXF`.

BASEMENT SCHEDULE OF DIMENSIONS & AREAS

Space / Zone	Dimensions (W × L)	Area
Car Ramp (14% Slope)	3.00m × 9.50m	28.50 m²
6 Parking Bays (Cars 1–6)	6 × (2.50m × 5.00m)	75.00 m²
Column-Free Turning Aisle	6.00m × 10.60m (R=5.5m)	63.60 m²
Stair + Elevator Core	4.20m × 3.80m	16.00 m²
Elec / Water / Storage	3.50m × 3.80m	13.30 m²
Private Gym	5.00m × 4.20m	21.00 m²
2× Baths + Kitchenette + Bins	5.00m × 4.40m	22.00 m²
Family Sitting Lounge (Pool View)	11.00m × 4.40m	48.40 m²
Total Basement Envelope	**16.00m × 19.00m**	**304.00 m²**
Independent Swimming Pool	**6.00m × 3.00m**	**18.00 m²**

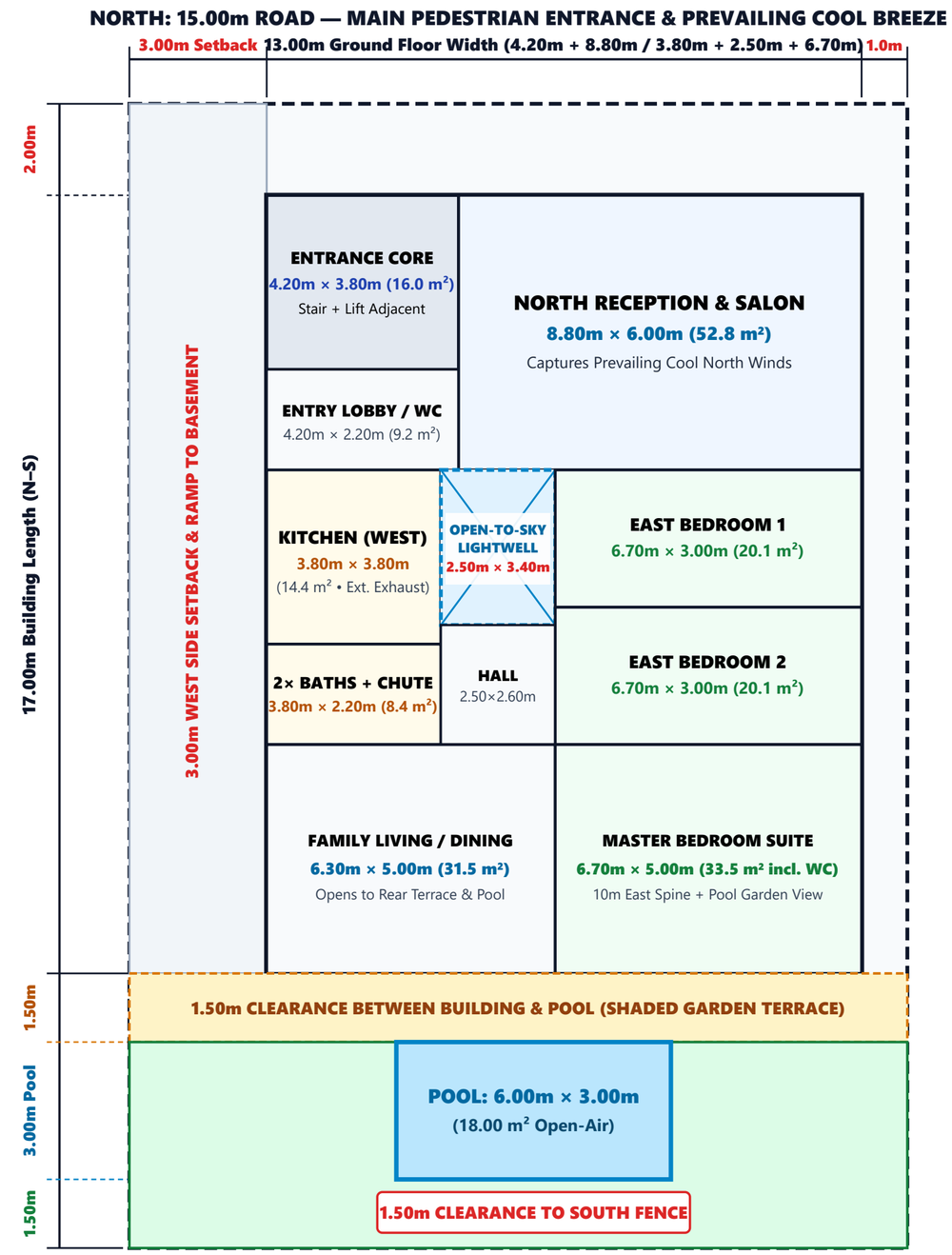
STRUCTURAL INDEPENDENCE OF POOL

- Pool shell rests on an independent RC raft foundation separated from the basement retaining wall by a **1.50m engineered granular backfill + XPS thermal/waterproof barrier**.
- Eliminates hydrostatic surcharge on basement walls while allowing full-height glazing for the Basement Family Lounge.

SHEET A-102: GROUND FLOOR PLAN (+0.60m) — EXACT 50.0% BUILT FOOTPRINT (212.50 m²)

17.00m × 25.00m Plot • 3.00m West Setback • 6.00m Rear Pool Garden (1.50m + 3.00m Pool + 1.50m) • Open-to-Sky Lightwell Clarified

REV 2 — ALL DIMENSIONS SHOWN



GROUND FLOOR NET BUILT AREA = 221.00 m² − 8.50 m² (Open Lightwell) = 212.50 m² (EXACT 50.0%)

ANSWER: WHAT IS THE WIND COURT / SOLAR CHIMNEY?

- **Yes — it is an Open-to-Sky Court / Lightwell (منور سماوي مفتوح):** An unroofed vertical open shaft measuring **2.50m × 3.40m (8.50 m²)** running continuously from Ground Floor up through the Roof.
- **How it works passively:** Cool prevailing Northern breezes enter the North Reception windows, flow across the apartment, and warm indoor air rises naturally up this central open shaft (*Solar Chimney / Stack Effect*) and exits through louvered roof vents.
- **Code & Area Benefit:** Because this **8.50 m²** shaft is open to the sky (unroofed), subtracting it from the `13.00m × 17.00m = 221.00 m²` outer wall rectangle gives **exact 212.50 m² (50.0% of 425 m²)** net built footprint while giving natural daylight and cross-ventilation to the central hall, kitchen, and bathrooms!

GROUND FLOOR ROOM DIMENSIONS (50% = 212.50 M²)

Room / Zone	Dimensions (W × L)	Net Area
Entrance Stair + Lift Core	4.20m × 3.80m	16.00 m²
Entry Lobby & Guest WC	4.20m × 2.20m	9.24 m²
North Reception & Salon	8.80m × 6.00m	52.80 m²
West Kitchen (Ext. Wall)	3.80m × 3.80m	14.44 m²
2× Bathrooms + Waste Chute	3.80m × 2.20m	8.36 m²
Central Distribution Hall	2.50m × 2.60m	6.50 m²
East Bedroom 1	6.70m × 3.00m	20.10 m²
East Bedroom 2	6.70m × 3.00m	20.10 m²
Master Suite 3 (SE + Bath)	6.70m × 5.00m	33.50 m²
Family Living / Garden Dining	6.30m × 5.00m	31.46 m²
Deduct Open-to-Sky Lightwell	*2.50m × 3.40m (Open)*	*(−8.50 m²)*
TOTAL GROUND FLOOR BUILT	**Exact 50.0% of 425 m²**	**212.50 m²**

SETBACKS: NORTH 2.00m • WEST 3.00m • EAST 1.00m • SOUTH 6.00m (1.50m + 3.00m POOL + 1.50m)

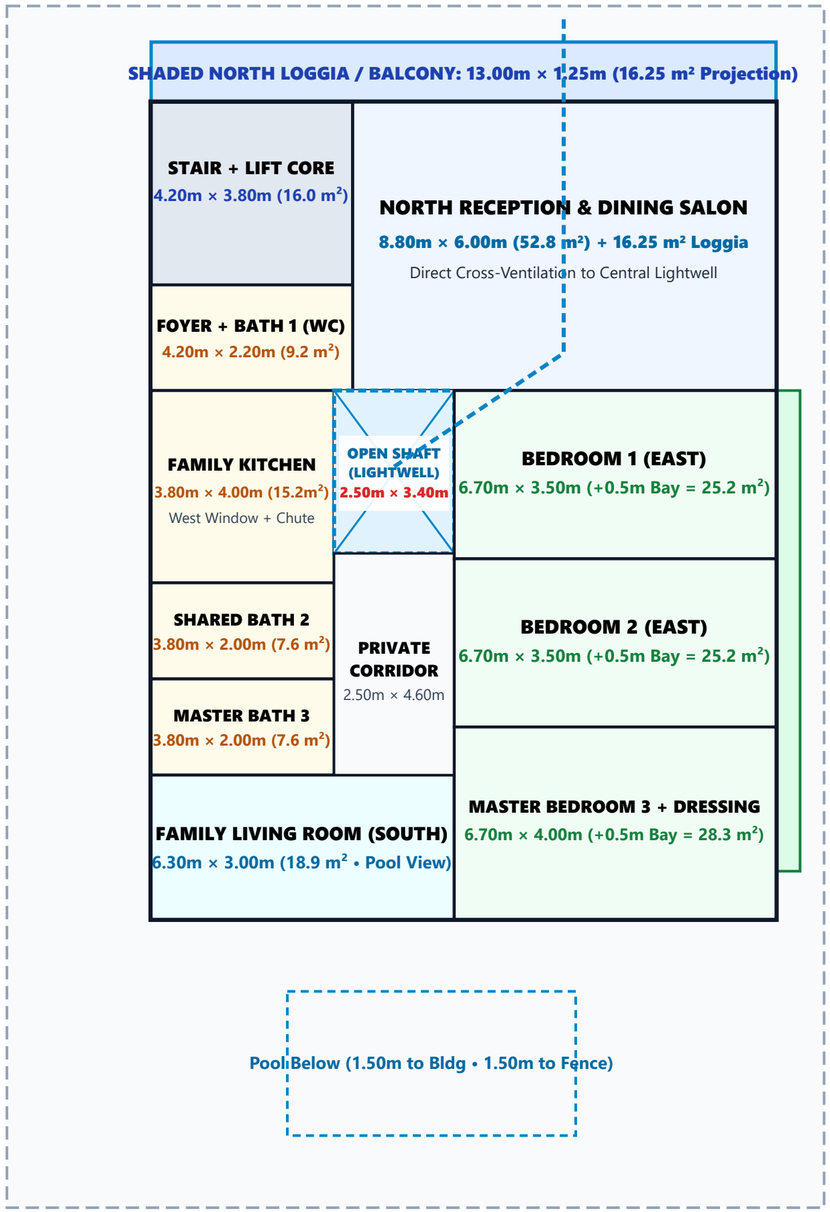
SHEET A-103: TYPICAL FLOORS 1, 2 & 3 — EXACT 55.0% BUILT AREA (233.75 m² / FLOOR)

1 Apartment Per Floor • North Reception (Prevailing Wind) • 3 East Bedrooms (10m Spine) • West Kitchen & 3 Bathrooms

REV 2 — FULL ROOM DIMENSIONS

NORTH: 15.00m ROAD — SHADED NORTH WIND LOGGIA (+1.25m CANTILEVER PROJECTION)

3.00m West **13.00m Core Envelope + 0.50m East Bay Window (Total 233.75 m² Net)**



TYPICAL FLOOR NET BUILT AREA = 212.50 + 16.25 (N. Loggia) + 5.00 (E. Bay) = 233.75 m² (EXACT 55.0%)

PASSIVE ZONING COMPLIANCE (TYPICAL FLOOR)

- **North Side (15m Road):** Grand Reception & Dining Salon (`8.80m × 6.00m`) + shaded `1.25m` North Loggia directly capture cool Etesian breezes.
- **East Side (10m Pedestrian Walkway):** All **3 Bedrooms** (`6.70m × 3.50m / 4.00m`) face the quiet pedestrian spine for gentle morning sunlight and zero street noise.
- **West External Wall: Kitchen** (`3.80m × 4.00m`) and **all 3 Bathrooms** buffer the apartment against hot afternoon western sun while enjoying direct exterior windows & exhaust.
- **Central Open Lightwell (`2.50m × 3.40m`):** Guarantees 100% natural cross-ventilation and daylight in the heart of every floor.

TYPICAL FLOOR SCHEDULE (FLOORS 1, 2 & 3)

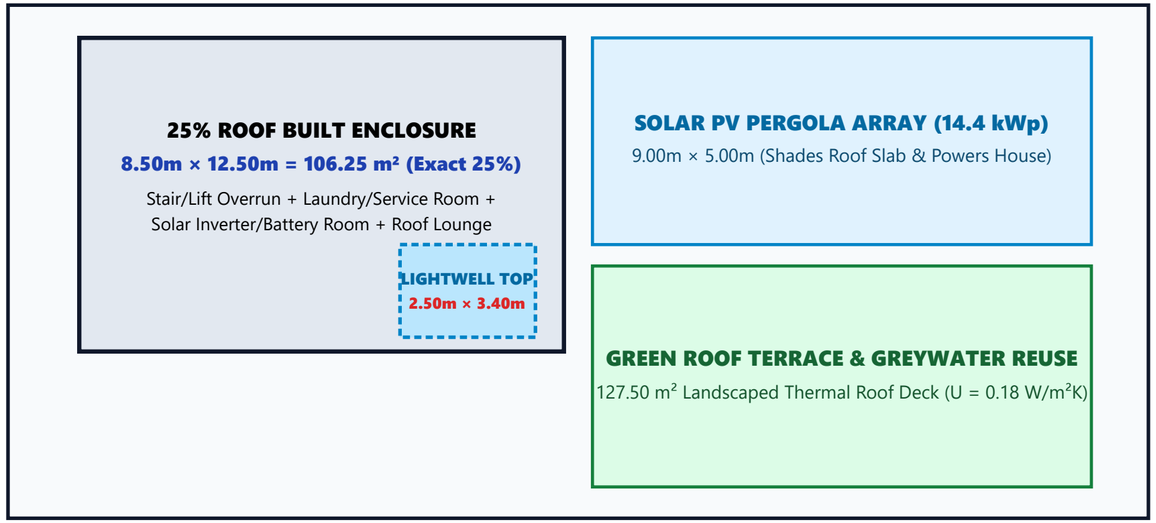
Room / Space	Dimensions (W × L)	Net Area
Stair + Elevator Core	4.20m × 3.80m	16.00 m²
Foyer + Guest Bath 1 (West)	4.20m × 2.20m	9.24 m²
North Reception & Dining	8.80m × 6.00m	52.80 m²
North Shaded Loggia	13.00m × 1.25m	16.25 m²
Family Kitchen (West Wall)	3.80m × 4.00m	15.20 m²
Bath 2 & Master Bath 3 (West)	2 × (3.80m × 2.00m)	15.20 m²
Bedroom 1 (East Spine)	6.70m × 3.50m + Bay	25.20 m²
Bedroom 2 (East Spine)	6.70m × 3.50m + Bay	25.20 m²
Master Bedroom 3 (SE)	6.70m × 4.00m + Bay	28.30 m²
Family Living + Corridor	6.30×3.00 + 2.50×4.60m	30.36 m²
Open-to-Sky Lightwell	*2.50m × 3.40m (Unroofed)*	*Excluded*
TOTAL PER TYPICAL FLOOR	**Exact 55.0% of 425 m²**	**233.75 m²**

SHEET A-104: ROOF PLAN (25% = 106.25 m²), SECTION A–A, CORE EVALUATION & ZERO-WASTE

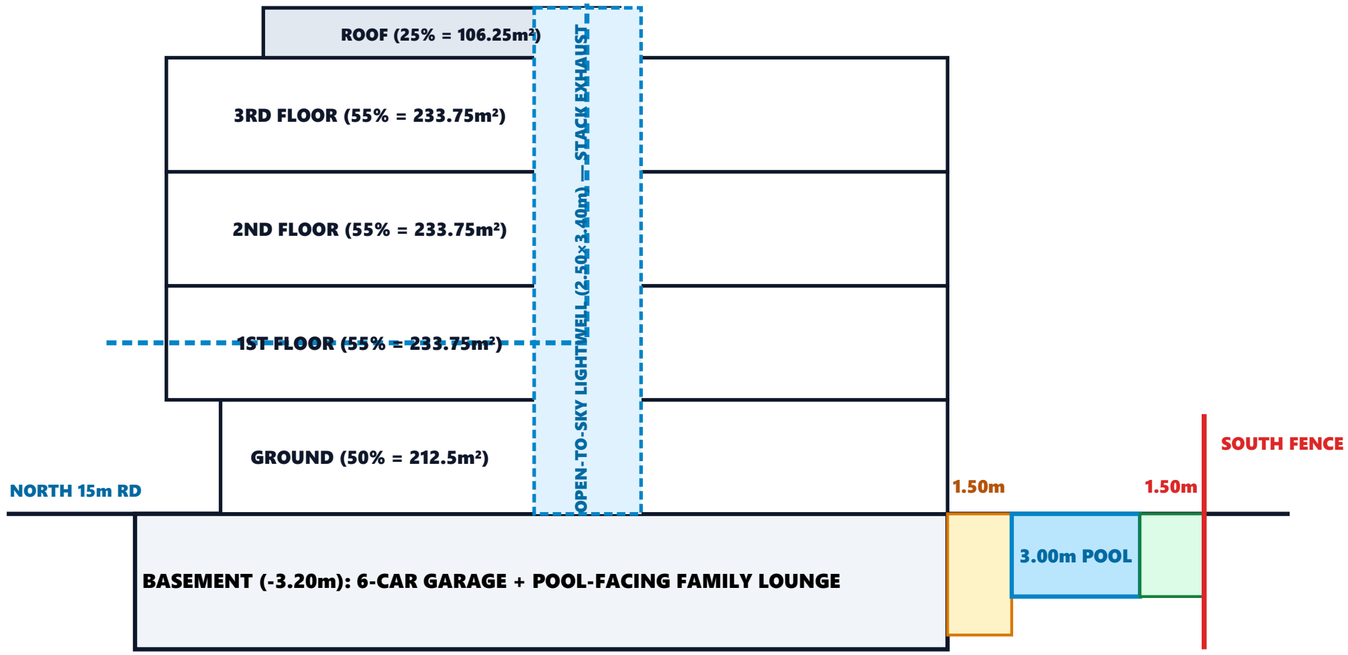
Longitudinal Section Showing Open-to-Sky Ventilation Shaft (Stack Effect) & Exact 1.50m + 3.00m + 1.50m Pool Clearances

REV 2 — ENGINEERING & SECTION

A. ROOF FLOOR PLAN (+13.80m) — EXACT 25.0% BUILT = 106.25 m² (8.50m × 12.50m)



B. LONGITUDINAL SECTION A–A (NORTH TO SOUTH — STACK VENTILATION & POOL CLEARANCES)



1.50m (Bldg–Pool) + 3.00m Pool + 1.50m (Pool–Fence) = 6.00m

CORE EVALUATION: WRAPPED VS. ADJACENT

Metric	Stair Wrapping Lift	Adjacent (Selected)
Area / Floor	22.60 m²	**16.00 m² (−6.6 m²/flr)**
4-Floor Saving	Baseline	**+26.4 m² usable living**
Structure	Open U-shaft	**Closed RC shear core**
Garage Aisle	Intrudes into turn	**Preserves 6.0m aisle**

ZERO-WASTE & SUSTAINABLE SYSTEMS

- **1.20m / 0.60m Modular Grid:** Eliminates blockwork & porcelain tile cutting waste (<1.5% site off-cuts).
- **4-Stream Sorted Waste Chute:** Located on West kitchen/service wall on every floor (Organic, Recyclables, Paper, Residual) feeding basement bins & composter.
- **Greywater & PV Loop:** Shower/laundry greywater filtration irrigates the South pool garden & flushes WCs (55% water savings); 14.4 kWp roof solar pergola offsets daytime AC load.

MASTER AREA SUMMARY (PLOT = 425.00 M²)

Level	Required Ratio	Designed Net Area
Basement (-3.20m)	6 Cars + Lounge	**304.00 m²**
Ground Floor	**50.0%**	**212.50 m² (Exact)**
Typical Floors (×3)	**55.0% / floor**	**233.75 m² × 3**
Roof Floor	**25.0%**	**106.25 m² (Exact)**
TOTAL BUILT GFA	**100% Compliant**	**1,324.00 m²**